SENTIMENT ANALYSIS FOR MENTAL HEALTH MONITORING USING MACHINE LEARNING

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Thesis Submitted in Partial Fulfillment of the requirement for the Degree of Bachelor of Studies in Computer Science

DEPARTMENT OF COMPUTER SCIENCE

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SESSION (2021-25)

DECLARATION

I, hereby declare that this thesis titled "**SENTIMENT ANALYSIS FOR MENTAL HEALTH MONITORING USING MACHINE LEARNING**" is the result of our research work. It has not been submitted previously, in whole or in part, for any other degree or diploma at any university or institution. And no part has been copied from any published source (except the references, standard, mathematical or geometrical models/equations/formulae/protocols etc.).

All sources of information used in this research have been duly acknowledged and referenced in accordance with academic standards.

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Approval Certificate

This is to certify that the research work presented in this thesis, entitled" **SENTIMENT ANALYSIS FOR MENTAL HEALTH MONITORING USING MACHINE LEARNING** " is conducted by **Maaz Kareem** under the supervision of **Dr. Hameed Hussain**.

No part of this thesis has been submitted anywhere else for any other degree. This thesis is submitted to the Department of Computer Science, University of Buner in partial fulfilment of the requirements for the degree of Bachelor of Studies in Computer Science.

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DEDICATION

I dedicate this research thesis to my beloved family, whose unwavering support, encouragement, and prayers have been my guiding light throughout my academic journey.

To my parents, for their endless love, sacrifices, and belief in my abilities; to my siblings, for their constant motivation and understanding; and to all my teachers and mentors, for inspiring me to strive for excellence.

This work is a testament to your faith in me, and I am forever grateful for your guidance and support.

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Chapter 1

Introduction

1.1 Background

In recent years, social media platforms have evolved into powerful channels for individuals to share their daily experiences, emotions, and personal thoughts. Millions of posts shared on platforms such as Twitter, Reddit, and Facebook contain linguistic patterns that reflect people's psychological states. These digital expressions can serve as valuable indicators of mental health conditions such as depression, anxiety, stress, personality disorder etc. Traditional methods for diagnosing mental health disorders typically rely on clinical assessments, self-reported questionnaires, or interviews conducted by professionals. While effective, these approaches are time-consuming, resource-intensive, and not always capable of capturing real-time emotional variations across large populations.

The emergence of artificial intelligence (AI) and natural language processing (NLP) has introduced new possibilities for analyzing mental health through text. By applying machine learning (ML) algorithms to social media data, it becomes possible to automatically detect linguistic indications associated with emotional distress or psychological disorders. Machine learning techniques such as Logistic Regression, Support Vector Machines (SVM), Random Forest, and Light Gradient Boosting Machine (LightGBM) have shown promising results in identifying mental health issues from textual information. These models rely on structured representations of text, such as Term Frequency–Inverse Document Frequency (TF-IDF) features, enabling them to classify posts based on emotional tone and content.

Such computational approaches can contribute significantly to early detection and mental health awareness. They offer scalable and data-driven solutions that complement traditional diagnosis, enabling researchers and doctors to monitor online behavioral trends and identify individuals who might require psychological support. Hence, sentiment analysis for mental health monitoring has emerged as a critical interdisciplinary research area combining data science, psychology, and computational linguistics.

1.2 Problem Statement

Mental health disorders such as depression, anxiety, bipolar, Suicidal Ideation Ideation, stress and personality disorder have become major global health concerns, affecting millions of individuals and contributing significantly to reduced quality of life and productivity. Despite the increasing occurrence of these conditions, timely detection and intervention remain challenging. Traditional clinical assessments rely heavily on self-reporting and professional evaluation, which are often limited by accessibility, cost, and social stigma. As a result, many individuals suffering from psychological distress remains undiagnosed or untreated.

At the same time, the widespread use of social media provides an opportunity to observe linguistic and behavioral indicators of mental health on a large scale. People frequently express their emotions, frustrations, and thoughts online, unknowingly leaving behind digital traces of their psychological state. However, extracting meaningful insights from such unstructured text data requires advanced computational techniques.

While deep learning models have gained attention for text classification tasks, they often demand extensive computational resources and large datasets. In contrast, machine learning (ML) methods such as Logistic Regression, Support Vector Machine (SVM), Random Forest, and LightGBM offer strong performance with greater interpretability and efficiency—especially when working with medium-sized datasets.

The primary problem addressed in this research is to develop an effective and interpretable machine learning-based system capable of analyzing social media text to detect mental health conditions. This research seeks to explore the extent to which classical ML models can classify user posts into categories such as Depression, Anxiety, Stress, Bipolar, Suicidal Ideation Ideation, Personality disorder and Normal while maintaining a balance between accuracy, interpretability, and computational efficiency.

1.3 Research Objectives

The primary objective of this research is to design and implement a machine learning-based framework capable of detecting and classifying mental health conditions from social media text data. The study aims to explore the potential of classical ML algorithms to accurately identify signs of psychological distress by analyzing linguistic patterns and sentiment expressions.

To achieve this overarching goal, the following specific objectives have been defined:

1. To collect and preprocess social media text data which is basically the social media post of the users, ensuring the removal of noise, duplicates, and irrelevant data for accurate analysis.
2. To extract meaningful textual features using natural language processing techniques such as tokenization, stop-word removal, lemmatization, and Term Frequency–Inverse Document Frequency (TF-IDF) vectorization.
3. To implement and evaluate multiple machine learning models—including Logistic Regression, Support Vector Machine (SVM), Random Forest, and LightGBM—for both binary and multi-class classification of mental health conditions.
4. To compare model performance based on evaluation metrics such as precision, recall, F1-score, and Area Under the Curve (AUC) to determine the most suitable algorithm for mental health detection.
5. To analyze and interpret key linguistic features influencing classification results, enhancing the interpretability and practical applicability of the developed system.

1.4 Research Questions

To guide the direction and scope of this study, the following research questions have been formulated:

1. How effectively can machine learning models classify mental health conditions such as Depression, Anxiety, Stress, Bipolar, Suicidal Ideation Ideation, Personality disorder and Normal based on textual data extracted from social media platforms that is Twitter, Reddit etc.?
2. Which machine learning algorithm among Logistic Regression, Support Vector Machine (SVM), Random Forest, and LightGBM achieves the best balance between accuracy, interpretability, and computational efficiency for mental health detection?
3. What linguistic and textual features play a significant role in identifying psychological distress in online user-generated content?
4. How does binary classification (Normal vs. Abnormal) compare to multi-class classification (Depression, Anxiety, Stress, Normal) in terms of performance and model interpretability?
5. Can machine learning-based sentiment analysis serve as a reliable tool to assist in the early detection and monitoring of mental health conditions on social media?

These research questions aim to establish a full understanding of how classical machine learning techniques can be leveraged to analyze linguistic patterns related to mental well-being and contribute toward scalable, interpretable, and resource-efficient mental health monitoring systems.

1.5 Scope and Significance of the Study

This research focuses on the application of machine learning (ML) techniques for the detection and classification of mental health conditions using textual data from social media posts. The study is limited to classical ML algorithms, specifically Logistic Regression, Support Vector Machine (SVM), Random Forest, and LightGBM, excluding deep learning models due to their higher computational resources demand and lower interpretability in small to medium datasets.

The two-dataset employed in this research is derived from publicly available sources, containing user-generated posts related to mental health discussions. The scope of the study includes both binary classification (Normal vs. Abnormal) and multi-class classification (Normal, Depression, Anxiety, Stress, Suicidal Ideation Ideation, Bipolar, and Personality disorder). Preprocessing techniques such as stop-word removal, tokenization, lemmatization, and TF-IDF vectorization are applied to ensure the textual data is clean, standardized, and suitable for model training and validation.

While the models developed in this study are not intended for clinical diagnosis, they can serve as a supportive tool for early detection, large-scale trend monitoring, and awareness generation regarding mental health conditions. The findings aim to assist researchers, psychologists, and policymakers in understanding how computational methods can complement traditional diagnostic approaches.

The significance of this research lies in the following aspects:

- It provides an understandable and resource-efficient framework for analyzing social media text to identify psychological suffering.
- It contributes to the growing body of research that bridges data science and mental health awareness, particularly within developing countries where clinical resources are limited.
- It demonstrates the practical potential of machine learning models to monitor mental health indicators in real time, fostering early intervention and support strategies.

1.6 Thesis Organization

This thesis is structured into five chapters, each addressing a specific component of the research process and contributing to the overall objectives of the study.

Chapter 1: Introduction

This chapter presents the background, problem statement, research objectives, research questions, scope, and significance of the study. It establishes the foundation for understanding the motivation behind using machine learning approaches for mental health monitoring through social media text analysis.

Chapter 2: Literature Review

This chapter provides an overview of previous studies and existing methodologies related to sentiment analysis, natural language processing, and machine learning applications in mental health detection. It critically analyzes the strengths and limitations of prior research, highlighting the research gap addressed in this study.

Chapter 3: Methodology

This chapter describes the research framework, including data collection, preprocessing, feature extraction, and the implementation of machine learning models. It also details the experimental setup, evaluation metrics, and procedures used to compare the performance of different algorithms.

Chapter 4: Results and Discussion

This chapter presents the experimental results for both binary and multi-class classification tasks. It discusses the comparative performance of Logistic Regression, SVM, Random Forest, and LightGBM models based on precision, recall, F1-score, and AUC. The findings are interpreted in light of model interpretability and computational efficiency.

Chapter 5: Conclusion and Future Work

This chapter summarizes the key findings of the research, concludes the study, and provides recommendations for future work. It also discusses the practical implications and potential extensions of the proposed machine learning framework for real-world mental health applications.

Chapter 2

Literature Review

2.1 Introduction

The field of mental health detection using artificial intelligence has rapidly evolved due to the increasing availability of user-generated data on social media. Researchers have explored various computational techniques to analyze emotional expressions, linguistic prompts, and behavioral indicators of psychological distress. Among these, machine learning (ML) methods have emerged as efficient and interpretable tools for classifying mental health conditions from text.

This chapter reviews existing literature on sentiment analysis, natural language processing (NLP) for mental health monitoring, and the application of classical ML algorithms such as Logistic Regression, Support Vector Machines (SVM), Random Forest, and LightGBM. It also highlights key findings, challenges, and research gaps that motivate the present study.

2.2 Sentiment Analysis and Mental Health

Sentiment analysis, also known as opinion mining, is the computational process of identifying and categorizing opinions expressed in text to determine the writer's attitude toward a particular topic or emotion. It plays a central role in mental health detection, as individuals often express their emotional states through language.

Research by Guntuku et al. (2017) demonstrated that social media posts contain valuable linguistic markers that correlate with depression and anxiety symptoms. Similarly, De Choudhury et al. (2013) analyzed Twitter data to measure population-level depression indicators and found that linguistic and temporal patterns can effectively reflect emotional well-being.

These studies established the foundation for applying computational text analysis to psychological research. However, they also revealed challenges such as informal language use, sarcasm, and contextual ambiguity, which require robust NLP preprocessing and well-designed models for accurate interpretation.

2.3 Machine Learning Approaches for Mental Health Detection

Machine learning (ML) techniques have become essential for automatically identifying mental health conditions from textual data. These models learn from labeled datasets to recognize

patterns associated with psychological states such as depression, anxiety, stress etc. Unlike deep learning, which requires large datasets and substantial computational resources, classical ML algorithms are computationally efficient and more interpretable, making them well-suited for medium-sized datasets and academic research applications.

2.3.1 Logistic Regression

Logistic Regression (LR) is one of the most widely used baseline models for binary and multi-class text classification. It estimates the probability that a given text belongs to a particular class based on weighted features. Hosmer and Lemeshow (2000) emphasized that logistic regression provides valuable interpretability through its coefficients, allowing researchers to understand how specific linguistic features influence predictions.
In mental health detection, LR has been used effectively to classify posts as "depressed" or "non-depressed" using TF-IDF features. Its strength lies in its simplicity, efficiency, and interpretability, though it may struggle to capture complex non-linear relationships in text data.

2.3.2 Support Vector Machine (SVM)

Support Vector Machine (SVM) is a robust supervised learning algorithm that identifies an optimal hyperplane to separate classes in high-dimensional spaces. Cortes and Vapnik (1995) introduced Support Vector Machine (SVM) as a powerful classifier, and subsequent research demonstrated its effectiveness in the text-based sentiment analysis.
In mental health detection, SVMs are particularly effective when dealing with sparse data representations such as TF-IDF vectors. Studies by Helmy et al. (2024) and Cao et al. (2025) showed that SVM models achieved high accuracy in identifying depression-related language from Twitter and Reddit data. However, SVMs can be computationally expensive when dealing with large datasets or multi-class problems

2.3.3 Random Forest (RF)

Random Forest is an ensemble-based algorithm that constructs multiple decision trees and aggregates their results to improve accuracy and robustness. Breiman (2001) demonstrated that RF models perform well in handling noisy and imbalanced data, which is common in social media text.

In mental health research, Random Forest has been applied to identify important linguistic features such as “sad,” “alone,” and “tired,” which are strongly correlated with depressive symptoms. It also offers the advantage of feature importance scores, providing insights into which words contribute most to classification decisions. However, Random Forest models can sometimes overfit if not properly tuned.

2.3.4 Light Gradient Boosting Machine (LightGBM)

LightGBM is a high-performance gradient-boosting framework that builds trees sequentially to correct errors made by previous models. Developed by Ke et al. (2017), LightGBM is known for its efficiency and scalability in text-based applications. Recent studies, including Ding et al. (2025), have shown that LightGBM performs competitively with deep learning models in mental health detection tasks, offering a strong balance between predictive performance and computational efficiency. Its ability to handle large feature sets and provide feature importance ranking makes it particularly valuable for sentiment analysis.

2.3.5 Comparative Analysis

Comparative research, such as the work by Ding et al. (2025) in Scientific Reports, found that classical ML models (Logistic Regression, SVM, Random Forest, and LightGBM) achieved comparable F1-scores and AUCs to deep learning models like GRU and ALBERT when applied to mental health detection. The study concluded that ML models remain preferable when interpretability and resource efficiency are priorities. These findings support the motivation of this research—to rely solely on machine learning approaches for an interpretable and computationally practical solution for mental health monitoring.

2.4 Challenges and Research Gaps

Although machine learning models have shown promising results in detecting mental health conditions from social media text, several challenges and limitations persist that affect the accuracy, generalizability, and ethical reliability of these systems. Identifying these gaps is crucial to guide future research and to ensure the responsible use of AI in mental health applications.

2.4.1 Data Quality and Imbalance

One of the most significant challenges in mental health detection is the quality and balance of the dataset. Social media data often contain noise, irrelevant content, slang, sarcasm, and spelling variations that hinder effective preprocessing. Moreover, datasets are frequently imbalanced, with a uneven number of "normal" posts compared to those expressing mental distress. Such imbalance can bias models toward the majority class, reducing their ability to detect minority conditions like stress or Suicidal Ideation Ideation ideation. Techniques like resampling or class weighting are commonly applied, but they can sometimes lead to overfitting or loss of linguistic diversity.

2.4.2 Feature Representation and Context

Machine learning models depend heavily on feature-engineering methods such as TF-IDF or Bag-of-Words representations. While these methods capture word frequency and importance, they often fail to preserve semantic meaning and contextual relationships between words. For instance, expressions of sarcasm or idiomatic phrases related to mental health might be misinterpreted due to lack of contextual understanding. This limitation constrains the depth of analysis achievable through traditional ML models compared to modern contextual embeddings used in deep learning.

2.4.3 Interpretability vs. Performance Trade-Off

A consistent theme in the literature is the trade-off between model interpretability and predictive performance. While ML models like Logistic Regression and Random Forest are more interpretable, they may underperform on complex linguistic patterns. Conversely, more advanced models, though potentially more accurate, act as "black boxes," limiting the transparency needed for ethical mental health applications. The challenge, therefore, lies in achieving an optimal balance between accuracy, transparency, and ethical accountability.

2.4.4 Ethical and Privacy Concerns

The use of social media data for mental health analysis raises serious ethical and privacy considerations. Many datasets are derived from publicly available posts that may include sensitive or personally recognizable information. Using such data without explicit consent could posture privacy risks. Moreover, incorrect classification, such as labeling a healthy user as depressed can have harmful consequences if used in real-world applications. Researchers have highlighted the

importance of anonymization, informed consent, and responsible interpretation when handling mental health data.

2.4.5 Research Gaps and Novelty of This Study

Despite substantial progress, several key gaps remain in the current literature on AI-based mental-health detection:

- Lack of standardized, ethically curated, and multi-class datasets that comprehensively represent diverse mental-health conditions.
- Limited prior research exploring cross-domain generalization, where models are trained on one dataset and evaluated on an entirely independent corpus to assess robustness a gap this study directly addresses.
- Under-representation of interpretable machine-learning models compared with deep-learning approaches, despite their advantages in transparency and computational efficiency.
- Insufficient integration of explainable AI (XAI) frameworks to enhance interpretability and ethical accountability. SHAP (SHapley Additive exPlanations), LIME (Local Interpretable Model-agnostic Explanations)

To bridge these gaps, this research examines the cross-domain generalization capability of several machine-learning models by training and validating on a publicly available dataset and testing on a newly constructed multi-class dataset. This approach provides deeper insight into model robustness across domains and contributes to the development of reproducible, interpretable, and computationally efficient frameworks for sentiment-based mental-health classification on social-media data.

2.5 Summary

This chapter has reviewed the key concepts, methodologies, and prior research related to the use of machine learning for mental health detection based on textual data from social media platforms. The literature demonstrates that emotional and linguistic cues embedded in user-generated content can serve as valuable indicators of psychological conditions such as depression, anxiety, stress etc.

Classical machine learning algorithms including Logistic Regression, Support Vector Machine (SVM), Random Forest, and Light Gradient Boosting Machine (LightGBM) have consistently shown strong potential in text classification tasks. They offer a desirable balance between accuracy, interpretability, and computational efficiency, making them suitable for studies involving limited data or requiring transparency in decision-making.

However, despite notable advancements, several challenges remain. These include data imbalance, limited contextual understanding, and ethical concerns related to privacy and consent. Furthermore, most existing research either focuses on deep learning models or neglects interpretability, leaving a research gap for ML-based frameworks that are both effective and explainable.

The present study builds upon these insights by proposing a machine learning-based sentiment analysis framework for monitoring mental health conditions on social media. The next chapter discusses the research methodology, including dataset description, preprocessing techniques, feature extraction, model implementation, and evaluation metrics used in this study.

Chapter 3

Methodology

3.1 Introduction

This chapter outlines the methodological framework adopted in this research for detecting mental health conditions using machine learning (ML) techniques. The study follows a structured pipeline that includes data collection, preprocessing, feature extraction, model development, evaluation and cross-domain generalization. The methodology was designed to ensure interpretability, efficiency, and reproducibility while maintaining ethical integrity in data handling.

The overall approach involves two main experiments:

1. **Binary Classification** – distinguishing between Normal and Abnormal (Mental Health Issue) posts.
2. **Multi-Class Classification** – categorizing posts into Normal, Depression, Anxiety, Stress, Bipolar, Suicidal Ideation post and Personality disorder.

Each phase of the methodology is detailed in the following sections.

3.2 Research Framework

The research framework of this study was designed to develop, train, and evaluate a machine learning-based system capable of detecting and classifying mental health conditions from social media text. The framework ensures reproducibility, generalization, and interpretability, integrating multiple datasets and well-established machine learning methods.

The complete process consists of five major stages: data acquisition, preprocessing, feature extraction, model training, and evaluation. Figure 3.1 (to be inserted in the final thesis document) conceptually illustrates the workflow of the research methodology.

3.2.1 Stage 1 – Data Acquisition

The first stage involved the collection and integration of three datasets containing social media text related to mental health conditions.

1. Combined Data.csv — a large aggregated dataset sourced from multiple publicly available Kaggle datasets, used primarily for model training and evaluation.

2. Sentiment_Mental_Health_Dataset.csv — a newly constructed dataset developed to ensure cross-domain generalization, combining Reddit data from the Zenodo repository with additional posts collected via the Reddit API.
3. Final_Merged_Dataset_Cleaned.csv — a merged dataset that combines the above two, including 20% data overlap, designed to test the generalizability of models across diverse data sources.

This multi-dataset approach enhanced the robustness and reliability of the study, reducing dataset bias and ensuring that the machine learning models could perform well across varying linguistic and contextual data domains.

3.2.2 Stage 2 – Data Preprocessing

The collected textual data underwent extensive preprocessing, including cleaning, tokenization, lemmatization, stop-word removal, and label encoding. These steps standardized the data and reduced noise, ensuring that only relevant linguistic information was passed to the feature extraction stage.

3.2.3 Stage 3 – Feature Extraction

Feature extraction was performed using the Term Frequency–Inverse Document Frequency (TF-IDF) method. This approach transformed the textual posts into numerical feature vectors, capturing important word-level patterns while maintaining computational efficiency.

3.2.4 Stage 4 – Model Training and Validation

Four classical machine learning algorithms Logistic Regression, Support Vector Machine (SVM), Random Forest, and Light Gradient Boosting Machine (LightGBM) were implemented for both binary and multi-class classification tasks. Each model was trained and validated using cross-validation techniques to ensure fairness and reliability.

3.2.5 Stage 5 – Evaluation and Comparative Analysis

The models were evaluated using a range of metrics Accuracy, Precision, Recall, F1-score, and AUC-ROC across all three datasets. Additionally, feature importance analysis was conducted for

the Random Forest and LightGBM models to interpret which textual features contributed most to the classification results.

In summary, the research framework was constructed to ensure a balance between accuracy, interpretability, and generalization. By leveraging multiple curated datasets and classical machine learning models, this study provides a reproducible and ethically sound foundation for automated sentiment-based mental health detection.

3.3 Dataset Description

This research utilized three distinct datasets, each carefully curated and processed to enhance the reliability and generalization capability of the machine learning models. The datasets contain textual data related to various mental health conditions, collected primarily from social media platforms such as Reddit and Twitter, where users discuss personal experiences, emotions, and psychological states.

All datasets were pre-labeled and structured for supervised machine learning tasks. Each entry consists of a textual statement and its corresponding mental health status label, making the datasets suitable for both binary and multi-class classification tasks. The following subsections describe each dataset in detail.

3.3.1 Dataset 1 - Combined Data.csv

The "Combined Data.csv" dataset was used for training, validation, and testing of the machine learning models. It is a comprehensive collection of mental health–related statements aggregated from multiple publicly available sources on Kaggle**.**

Data Sources:

- 3k Conversations Dataset for Chatbot
- Depression Reddit Cleaned
- Human Stress Prediction
- Predicting Anxiety in Mental Health Data
- Mental Health Dataset Bipolar
- Reddit Mental Health Data

- Students Anxiety and Depression Dataset
- Suicidal Mental Health Dataset
- Suicidal Tweet Detection Dataset

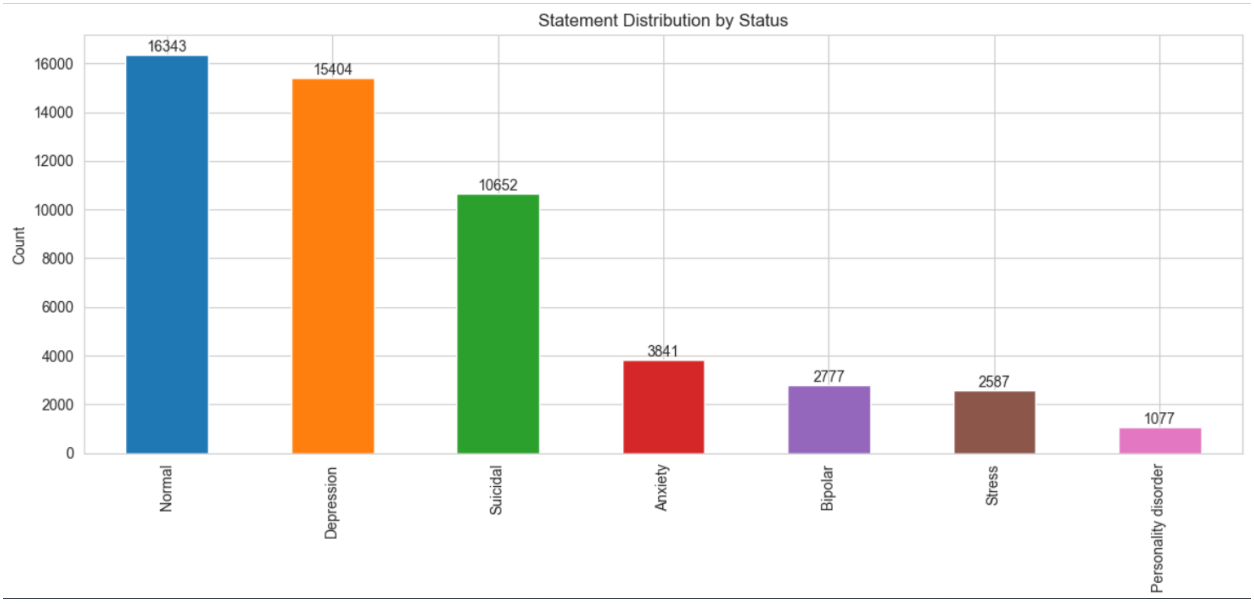

Figure 1

Data Overview:

This dataset contains posts categorized into seven mental health statuses: *Normal, Depression, Suicidal, Anxiety, Stress, Bipolar, and Personality Disorder.*

Structure:

- **unique_id** – unique identifier for each entry
- **statement** – text or post content
- **mental_health_status** – assigned mental health label

Purpose:

This dataset was primarily used for model training and evaluation due to its diversity and

representativeness, making it ideal for sentiment analysis, chatbot development, and research in computational mental health.

3.3.2 Dataset 2 - Sentiment_Mental_Health_Dataset.csv

The "Sentiment_Mental_Health_Dataset.csv" was a novel dataset constructed from scratch to achieve cross-domain generalization and enhance the robustness of the classification models.

Data Sources:

- Zenodo Reddit Mental Health Dataset (for Depression, Anxiety, Suicidal Ideation, Bipolar Disorder, and Normal)
- Reddit Data Collection via PRAW API (for Stress and Personality Disorder)

Data Overview:

This dataset includes seven categories:

*Depression, Anxiety, Suicidal Ideation, Bipolar Disorder, Normal, Stress, and Personality Disorder***.**

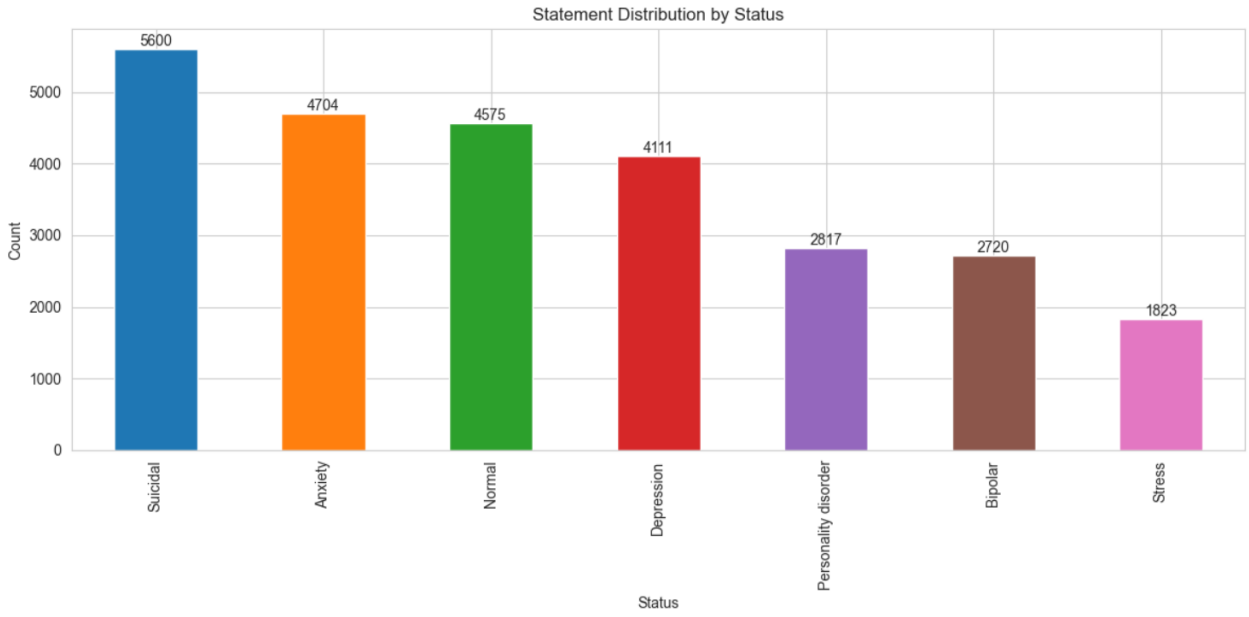

Figure 2

Data Cleaning and Processing:

Duplicate or empty posts were removed, and the remaining data were standardized by combining

titles and self-text fields into a unified textual corpus. The resulting dataset provides a balanced representation across all mental health categories.

Ethical Considerations:

All data were collected from publicly available Reddit posts. No usernames or identifiable information were included, ensuring full compliance with research ethics and privacy protection.

3.3.3 Dataset 3 - Final_Merged_Dataset_Cleaned.csv

The "Final_Merged_Dataset_Cleaned.csv" was created to evaluate the cross-domain performance of the trained models. It merges the complete Sentiment_Mental_Health_Dataset.csv with 20% of the data from Combined Data.csv, preserving essential linguistic diversity from both sources.

Data Sources:

- Sentiment_Mental_Health_Dataset.csv
- Combined Data.csv

Data Overview:

The merged dataset also contains seven mental health categories:
Normal, Depression, Suicidal, Anxiety, Stress, Bipolar, and *Personality Disorder.*

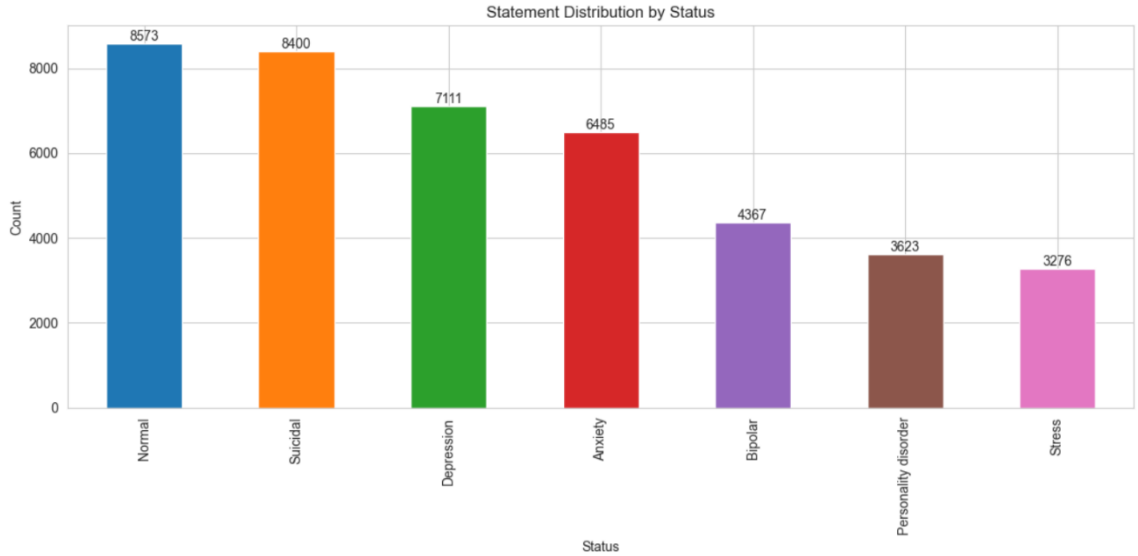

Figure 3

This merged dataset enabled the study to assess model generalization, ensuring that classification accuracy was not limited to a single data domain. It thus serves as a bridge between the two primary

datasets, enhancing the robustness and transferability of the models developed in this research. In summary, these three datasets collectively provided a diverse, ethically curated, and comprehensive foundation for developing and evaluating machine learning models for sentiment-based mental health detection. Each dataset contributed uniquely to training, validation, and cross-domain testing, ensuring both reliability and generalizability of the research findings.

3.4 Data Preprocessing

Before applying any machine learning algorithms, the collected text data underwent a comprehensive preprocessing phase to ensure quality, consistency, and suitability for model training. Social media text is often informal, containing noise such as hashtags, URLs, emojis, and grammatical inconsistencies. Therefore, preprocessing is essential to remove irrelevant elements and transform raw text into a standardized format suitable for computational analysis.

The preprocessing pipeline implemented in this study consisted of the following major steps:

3.4.1 Text Cleaning

All unnecessary and non-linguistic elements were removed from the dataset, including:

- URLs, mentions (e.g., "@user"), and hashtags (#).
- Punctuation marks, numbers, and special characters.
- Extra whitespace and non-alphabetic symbols.

This step ensured that the text contained only meaningful linguistic content relevant to sentiment and emotion detection.

3.4.2 Tokenization

Each sentence was split into individual words or tokens using Python's **NLTK** library. Tokenization allows the text to be processed at the word level, enabling feature extraction techniques such as TF-IDF to analyze word frequency and distribution patterns.

3.4.3 Stop-word Removal

Commonly used words such as "the," "is," "and," and "was" that do not contribute to sentiment meaning were removed from the corpus. Stop-word removal reduced data dimensionality and improved the learning efficiency of machine learning models.

3.4.4 Lemmatization

Each token was converted into its base or dictionary form (lemma) using lemmatization. For example, “running,” “runs,” and “ran” were all reduced to “run.” This ensured that different morphological variants of a word were treated uniformly, improving feature consistency and model accuracy.

3.4.5 Label Encoding

Textual labels (e.g., “Depression,” “Anxiety,” “Stress,” “Normal” “Bipolar,” “Personality disorder,” “Suicidal ideation,”) were converted into numerical form using label encoding to make them compatible with machine learning algorithms.

3.4.6 Train-Test Split

Finally, the first dataset was divided into training, testing and validation subsets, typically in a 60:20:20 ratio. The training set was used to build the model, while the validation set is used for tuning and test set is for evaluation. The next two datasets are divided 80:20 and its test set is used for evaluating generalizations. This preprocessing workflow ensured that the text data was clean, standardized, and properly structured for feature extraction and subsequent model training.

3.5 Feature Extraction

After data preprocessing, the next crucial step was to transform textual information into a numerical representation that could be effectively processed by machine learning algorithms. Since models cannot interpret raw text directly**,** feature extraction techniques are applied to convert words and sentences into structured numerical vectors while preserving their semantic significance.

In this study, the Term Frequency–Inverse Document Frequency (TF-IDF) method was employed for feature extraction, owing to its simplicity, efficiency, and effectiveness in representing textual data in sentiment and classification tasks.

3.5.1 Term Frequency–Inverse Document Frequency (TF-IDF)

TF-IDF is a statistical technique that evaluates how important a word is to a document in a collection or corpus. It combines two components:

- **Term Frequency (TF):** Measures how frequently a word appears in a document.

- **Inverse Document Frequency (IDF):** Measures how unique or rare a word is across all documents in the corpus.

The TF-IDF score for a term t in a document d is calculated as:

$$TF - IDF\ (t, d) = TF(t, d) \times \log\left(\frac{N}{DF(t)}\right)$$

where:

- $TF(t, d)$ is the frequency of term t in document d,
- (N) is the total number of documents, and
- $DF(t)$ is the number of documents containing the term t.

Words with higher TF-IDF scores are considered more relevant to the document's meaning. This representation helps distinguish emotionally charged words (e.g., "sad," "tired," "hopeless") from common ones (e.g., "the," "and," "was").

3.5.2 Implementation Details

The **TF-IDF vectorizer** from the scikit-learn library was used with the following configurations:

- **Maximum Features:** 10,000 (to limit dimensionality).
- **N-grams**: Unigrams and bigrams (to capture single words and short phrases).
- **Stop Words**: English stop words were removed automatically.

The output of the TF-IDF process was a sparse matrix, where each row represented a social media post, and each column corresponded to a unique term or n-gram feature. This matrix served as the input for all subsequent machine learning models.

3.6 Machine Learning Models

To identify and classify mental health conditions based on social media text, several supervised machine learning algorithms were implemented and compared. These models were selected for their proven performance in text classification tasks, interpretability, and computational efficiency. The selected algorithms include Logistic Regression, Support Vector Machine (SVM), Random Forest, and Light Gradient Boosting Machine (LightGBM).

Each algorithm was trained using the TF-IDF features derived from the preprocessed dataset and evaluated under identical conditions for both binary and multi-class classification experiments.

3.6.1 Logistic Regression (LR)

Logistic Regression is a probabilistic model that predicts the likelihood of an input belonging to a particular class. It uses the sigmoid function to map predictions between 0 and 1, making it suitable for binary and multi-class text classification tasks.

In this study, Logistic Regression served as a baseline model due to its simplicity and interpretability. The model's coefficients provide insights into how specific words influence the prediction of mental health categories (e.g., "sad" indicating depression). The implementation utilized L2 regularization to prevent overfitting and the 'lbfgs' optimizer for efficient convergence.

3.6.2 Support Vector Machine (SVM)

Support Vector Machine (SVM) is a powerful algorithm designed to find the optimal hyperplane that separates classes in a high-dimensional feature space. By maximizing the margin between classes, SVM achieves strong generalization performance, especially for text data represented by sparse TF-IDF vectors.

In this research, the Linear SVM (using the LinearSVC implementation from scikit-learn) was employed. It demonstrated robustness in handling overlapping linguistic features between different mental health conditions. SVM is particularly effective when there is limited training data but high-dimensional features, as is common in natural language processing tasks.

3.6.3 Random Forest (RF)

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting. Each tree is trained on a random subset of data and features, and the final prediction is determined through majority voting.

The Random Forest Classifier was chosen for its ability to handle noisy and nonlinear relationships in the dataset. Additionally, the model provides feature importance scores, which help identify the most influential words contributing to mental health classification. The parameters were optimized using cross-validation, focusing on the number of estimators and maximum tree depth.

3.6.4 Light Gradient Boosting Machine (LightGBM)

LightGBM is a gradient boosting framework that builds decision trees sequentially, where each new tree aims to correct the errors made by previous ones. It is optimized for both speed and accuracy, making it suitable for large-scale text classification tasks.

In this study, LightGBM was implemented due to its superior computational efficiency and performance stability. It also supports feature importance analysis, enabling the identification of key textual patterns associated with depression, anxiety, or stress. The model parameters were fine-tuned through grid search to achieve optimal accuracy and F1-score.

3.6.5 Model Training and Validation

All models were trained using the training subset (80%) and evaluated on the testing subset (20%) of the dataset. Hyperparameter optimization was performed using Grid Search Cross-Validation to ensure fair comparison across models.

Each algorithm was evaluated on both binary classification (Normal vs. Abnormal) and multi-class classification (Normal, Depression, Anxiety, Stress) experiments to assess generalizability and robustness.

3.7 Evaluation Metrics

To assess the performance of the machine learning models, several quantitative evaluation metrics were employed. These metrics measure the effectiveness of the models in correctly classifying mental health conditions and provide insights into their predictive reliability and generalization capability.

Since the dataset used in this research involves both binary and multi-class classification problems, multiple complementary metrics were considered. These include Accuracy, Precision, Recall, F1-Score, and the Area Under the Receiver Operating Characteristic Curve (AUC-ROC).

3.7.1 Accuracy

Accuracy measures the proportion of correctly classified instances among all instances in the dataset. Although it is a straightforward measure, it may not fully represent performance when the data is imbalanced.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives

3.7.2 Precision

Precision represents the proportion of correctly predicted positive cases among all predicted positives. It reflects how many identified “mental health” cases were actually correct.

$$Precision = \frac{TP}{TP + FP}$$

High precision indicates that the model makes few false-positive errors.

3.7.3 Recall (Sensitivity)

Recall measures the proportion of correctly predicted positive cases out of all actual positives. It represents the model's ability to identify individuals truly suffering from a mental health condition.

$$Recall = \frac{TP}{TP + FN}$$

A high recall value suggests that the model effectively minimizes false negatives, which is crucial in mental health applications where missing an actual case can have serious implications.

3.7.4 F1-Score

The F1-score provides a harmonic mean between precision and recall, offering a single balanced metric to evaluate performance. It is particularly valuable when dealing with imbalanced datasets.

$$F1 - Score = 2 * \frac{Precision * Recall}{Precision + Recall}$$

Higher F1-scores indicate that the model achieves a strong balance between correctly identifying positive cases and minimizing false alarms.

3.7.5 Area Under the Curve (AUC-ROC)

The AUC-ROC evaluates the model's ability to distinguish between classes by plotting the True Positive Rate (TPR) against the False Positive Rate (FPR) at various threshold settings. A model with an AUC close to 1.0 indicates excellent separability between classes, while a value near 0.5 suggests random guessing.

These metrics were computed for each model to comprehensively evaluate their classification performance. They provided the foundation for selecting the best-performing machine learning algorithm for sentiment-based mental health detection.

3.8 Experimental Setup

This section describes the technical configuration, experimental design, and validation strategy used to conduct all experiments in this research. The setup was designed to ensure that the results were reproducible, unbiased, and generalizable across multiple datasets. The experiments were performed in two phases — binary classification and multi-class classification — using all three datasets described earlier.

3.8.1 Hardware and Software Environment

All experiments were conducted on a personal computing environment suitable for academic-level research in machine learning and natural language processing. The specifications are as follows:

- Processor: Intel® Core™ i5 (12th Generation)
- RAM: 8 GB
- Operating System: Windows 11 (64-bit)
- Programming Language: Python 3.10
- Development Platform: Jupyter Notebook

Key Libraries Used:

- *pandas*, *numpy* — for data manipulation and analysis
- *nltk*, *spacy* — for natural language preprocessing
- *scikit-learn* — for model implementation, feature extraction, and evaluation
- *lightgbm* — for Light Gradient Boosting Machine (LightGBM)
- *matplotlib*, *seaborn* — for visualization and performance comparison

This environment provided sufficient computational capability to train and evaluate multiple models efficiently while maintaining interpretability and consistency.

3.8.2 Experimental Design

The research was divided into two primary experimental phases, applied across three datasets:

1. **Binary Classification Phase**

- Classes: *Normal* vs. *Abnormal (Mental Health Issue)*
- Objective: Detect whether a post indicates any psychological distress.
- Datasets Used: *Combined Data.csv*, *Sentiment_Mental_Health_Dataset.csv*, and *Final_Merged_Dataset_Cleaned.csv*.

2. Multi-Class Classification Phase

- Classes: Normal, Depression, Anxiety, Stress, Suicidal, Bipolar, and Personality Disorder
- Objective: Distinguish between specific mental health conditions.
- Datasets Used: All three datasets, with cross-domain testing using *Sentiment_Mental_Health_Dataset.csv* and *Final_Merged_Dataset_Cleaned.csv* to assess model generalization.

Each model was trained and evaluated on every dataset to determine its performance consistency across different data distributions and linguistic contexts.

3.8.3 Model Optimization and Validation

To ensure fair and optimal model comparison, hyperparameter tuning was performed using Grid Search Cross-Validation (GridSearchCV) for each machine learning algorithm. The key parameters optimized included:

- Logistic Regression: Regularization strength (C), solver type
- SVM: Kernel type, regularization parameter (C)
- Random Forest: Number of estimators, maximum depth, and splitting criterion
- LightGBM: Learning rate, number of leaves, and maximum tree depth

First dataset was split into training (60%) and testing (20%) subsets, and validation (20%) was carried out using five-fold cross-validation (k=5). This approach minimized data dependency and ensured robust, unbiased performance evaluation.

3.8.4 Cross-Domain Evaluation

A key feature of this study was cross-domain testing to assess model generalization. Models trained on the Combined Data.csv were tested on Sentiment_Mental_Health_Dataset.csv and Final_Merged_Dataset_Cleaned.csv, that how models preform on new unseen and mixed data. This evaluation provided insights into the ability of models to perform effectively when exposed to unseen data domains a critical factor for real-world deployment in diverse online environments.

3.8.5 Visualization and Interpretation

To interpret model behavior and feature influence, multiple visualization techniques were employed:

- Confusion matrices for class-wise performance evaluation.
- ROC curves for measuring separability between classes.
- Feature importance plots (for Random Forest and LightGBM) to highlight key linguistic indicators of mental health conditions.

These visualizations facilitated a deeper understanding of each model's decision-making process and interpretability, ensuring the study's findings are both transparent and meaningful. In summary, this experimental setup ensured methodological rigor, ethical integrity, and reproducibility. The integration of three diverse datasets allowed for robust model evaluation across domains, validating the effectiveness and generalizability of classical machine learning

3.9 Summary

This chapter has presented the methodological framework and experimental procedures adopted for detecting mental health conditions from social media text using machine learning techniques. The research methodology was structured into distinct phases—data collection, preprocessing, feature extraction, model implementation, and evaluation—to ensure systematic and reproducible results.

Text preprocessing steps such as cleaning, tokenization, lemmatization, and stop-word removal ensured that the data was clean and meaningful. The TF-IDF method was used to convert textual information into numerical form, enabling effective input representation for various machine learning models.

Four classical machine learning algorithms, Logistic Regression, Support Vector Machine (SVM), Random Forest, and Light Gradient Boosting Machine (LightGBM), were implemented and optimized through cross-validation. Their performance was evaluated using standard metrics including accuracy, precision, recall, F1-score, and AUC.

The experiments were conducted in two phases: binary classification and multi-class classification, to analyze model performance under different task complexities. The chosen experimental setup, combined with careful parameter tuning and feature interpretation, ensured a fair and robust comparison among the models.

The next chapter presents and discusses the results obtained from these experiments, providing a detailed analysis of model performance, comparative evaluation, and key findings derived from the study.

Chapter 4

Results and Discussion

4.1 Introduction

This chapter presents the experimental results and analysis of the machine learning models developed for sentiment-based mental health detection. The performance of each algorithm Logistic Regression, Support Vector Machine (SVM), Random Forest, and Light Gradient Boosting Machine (LightGBM) was evaluated across two tasks: binary classification (Normal vs. Abnormal) and multi-class classification (Normal, Depression, Anxiety, Stress, Bipolar, suicidal ideation and Personality disorder).

The results are analyzed using standard evaluation metrics, including accuracy, precision, recall, F1-score, and AUC. This chapter also discusses the comparative performance of models, identifies the best-performing approach, and interprets key linguistic features that contribute to mental health prediction.

4.2 Binary Classification Results

The binary classification experiment aimed to distinguish between *Normal* and *Abnormal* posts, where "Abnormal" represents any form of mental distress (Depression, Anxiety, Stress, Bipolar, suicidal ideation and Personality disorder).

4.2.1 SVM Classification Report

Table 1. Combined Data.csv

```
--- Evaluation Report: SVM ---

Classification Report:
              precision    recall  f1-score   support

           0       0.91      0.91      0.91      3308
           1       0.96      0.96      0.96      7229

    accuracy                           0.94     10537
   macro avg       0.93      0.93      0.93     10537
weighted avg       0.94      0.94      0.94     10537

Weighted F1 Score: 0.9415
95% CI for F1 Score: [0.9370, 0.9458]
AUC Score: 0.9816
95% CI for AUC: [0.9793, 0.9836]
```

Table 2. Sentiment_Mental_Health_Dataset.csv

```
--- Evaluation Report: SVM ---

Classification Report:
              precision    recall  f1-score   support

           0       0.91      0.91      0.91      3308
           1       0.96      0.96      0.96      7229

    accuracy                           0.94     10537
   macro avg       0.93      0.93      0.93     10537
weighted avg       0.94      0.94      0.94     10537

Weighted F1 Score: 0.9416
95% CI for F1 Score: [0.9372, 0.9457]
```

Table 3. Final_Merged_Dataset_Cleaned.csv

```
--- Evaluation Report: SVM ---

Classification Report:
              precision    recall  f1-score   support

           0       0.91      0.91      0.91      3308
           1       0.96      0.96      0.96      7229

    accuracy                           0.94     10537
   macro avg       0.93      0.93      0.93     10537
weighted avg       0.94      0.94      0.94     10537

Weighted F1 Score: 0.9417
95% CI for F1 Score: [0.9375, 0.9460]
```

4.2.2 Logistic Regression Classification Report

Table 4. Combined Data.csv

```
--- Evaluation Report: LR ---

Classification Report:
              precision    recall  f1-score   support

           0       0.90      0.90      0.90      3308
           1       0.95      0.95      0.95      7229

    accuracy                           0.94     10537
   macro avg       0.93      0.93      0.93     10537
weighted avg       0.94      0.94      0.94     10537

Weighted F1 Score: 0.9365
95% CI for F1 Score: [0.9314, 0.9409]
```

Table 5. Sentiment_Mental_Health_Dataset.csv

```
--- Evaluation Report: LR ---

Classification Report:
              precision    recall  f1-score   support

           0       0.90      0.90      0.90      3308
           1       0.95      0.95      0.95      7229

    accuracy                           0.94     10537
   macro avg       0.93      0.93      0.93     10537
weighted avg       0.94      0.94      0.94     10537

Weighted F1 Score: 0.9364
95% CI for F1 Score: [0.9319, 0.9413]
```

Table 6. Final_Merged_Dataset_Cleaned.csv

```
--- Evaluation Report: LR ---

Classification Report:
              precision    recall  f1-score   support

           0       0.90      0.90      0.90      3308
           1       0.95      0.95      0.95      7229

    accuracy                           0.94     10537
   macro avg       0.93      0.93      0.93     10537
weighted avg       0.94      0.94      0.94     10537

Weighted F1 Score: 0.9363
95% CI for F1 Score: [0.9320, 0.9408]
```

4.2.3 Random Forest Classification Report

Table 7. Combined Data.csv

```
--- Evaluation Report: Random Forest ---

Classification Report:
              precision    recall  f1-score   support

           0       0.91      0.88      0.89      3308
           1       0.94      0.96      0.95      7229

    accuracy                           0.94     10537
   macro avg       0.93      0.92      0.92     10537
weighted avg       0.93      0.94      0.93     10537

Weighted F1 Score: 0.9347
95% CI for F1 Score: [0.9300, 0.9390
```

Table 8. Sentiment_Mental_Health_Dataset.csv

```
--- Evaluation Report: Random Forest ---

Classification Report:
              precision    recall  f1-score   support

           0       0.91      0.88      0.89      3308
           1       0.94      0.96      0.95      7229

    accuracy                           0.94     10537
   macro avg       0.93      0.92      0.92     10537
weighted avg       0.93      0.94      0.93     10537

Weighted F1 Score: 0.9347
95% CI for F1 Score: [0.9299, 0.9394]
```

Table 9. Final_Merged_Dataset_Cleaned.csv

```
--- Evaluation Report: Random Forest ---

Classification Report:
              precision    recall  f1-score   support

           0       0.91      0.88      0.89      3308
           1       0.94      0.96      0.95      7229

    accuracy                           0.94     10537
   macro avg       0.93      0.92      0.92     10537
weighted avg       0.93      0.94      0.93     10537

Weighted F1 Score: 0.9346
95% CI for F1 Score: [0.9300, 0.9391]
```

4.2.4 LightGBM Classification Report

Table 10. Combined Data.csv

```
--- Evaluation Report: LGBM ---

Classification Report:
              precision    recall  f1-score   support

           0       0.90      0.90      0.90      3308
           1       0.95      0.95      0.95      7229

    accuracy                           0.94     10537
   macro avg       0.93      0.93      0.93     10537
weighted avg       0.94      0.94      0.94     10537

Weighted F1 Score: 0.9363
95% CI for F1 Score: [0.9314, 0.9408]
```

Table 11. Sentiment_Mental_Health_Dataset.csv

```
--- Evaluation Report: LGBM ---

Classification Report:
              precision    recall  f1-score   support

           0       0.90      0.90      0.90      3308
           1       0.95      0.95      0.95      7229

    accuracy                           0.94     10537
   macro avg       0.93      0.93      0.93     10537
weighted avg       0.94      0.94      0.94     10537

Weighted F1 Score: 0.9364
95% CI for F1 Score: [0.9319, 0.9410]
```

Table 12. Final_Merged_Dataset_Cleaned.csv

```
--- Evaluation Report: LGBM ---

Classification Report:
              precision    recall  f1-score   support

           0       0.90      0.90      0.90      3308
           1       0.95      0.95      0.95      7229

    accuracy                           0.94     10537
   macro avg       0.93      0.93      0.93     10537
weighted avg       0.94      0.94      0.94     10537

Weighted F1 Score: 0.9362
95% CI for F1 Score: [0.9315, 0.9411]
```

Analysis

All models demonstrated strong predictive capability in distinguishing between normal and mentally distressed posts.

- **SVM** achieved the highest performance across all metrics, showing excellent generalization and separability (F1 Score = 0.9364, AUC = 0.9816).
- **LightGBM** followed closely, confirming its robustness for high-dimensional textual data.
- **Logistic Regression** performed reliably as a baseline model,

- **Random Forest** showed slightly lower accuracy due to its sensitivity to feature sparsity in TF-IDF data.

The results confirm that classical ML models, when properly tuned, can perform remarkably well for binary sentiment-based mental health classification without requiring deep learning architectures.

4.3 Multi-Class Classification Results

In the multi-class classification experiment, the models were trained to categorize posts into four distinct classes: Depression, Anxiety, Stress, Bipolar, suicidal ideation and Personality disorder**.** This task is inherently more complex than binary classification because of overlapping linguistic expressions and subtle emotional variations across mental health conditions.

The comparative performance of all models on the multi-class dataset is summarized below.

4.3.1 SVM Classification Report

Table 13. Combined Data.csv

```
--- Evaluation Report: SVM ---

Classification Report:
              precision    recall  f1-score   support

           0     0.8276    0.7947    0.8108       755
           1     0.9061    0.6774    0.7752       527
           2     0.6678    0.7444    0.7040      3016
           3     0.8576    0.9429    0.8982      3308
           4     0.9224    0.4515    0.6062       237
           5     0.7554    0.4608    0.5724       536
           6     0.6959    0.6372    0.6652      2158

    accuracy                         0.7640     10537
   macro avg     0.8047    0.6727    0.7189     10537
weighted avg     0.7667    0.7640    0.7593     10537

Weighted F1 Score: 0.7593
95% CI for F1 Score: [0.7505, 0.7673]
```

Table 14. Sentiment_Mental_Health_Dataset.csv

```
--- Evaluation Report: SVM ---

Classification Report:
              precision    recall  f1-score   support

           0     0.7955    0.2778    0.4118       882
           1     0.9205    0.5167    0.6619       538
           2     0.2239    0.8413    0.3536       832
           3     0.5326    0.5704    0.5509       959
           4     1.0000    0.0157    0.0308       575
           5     0.8235    0.2932    0.4324       382
           6     0.7313    0.2396    0.3609      1102

    accuracy                         0.4089      5270
   macro avg     0.7182    0.3935    0.4003      5270
weighted avg     0.6811    0.4089    0.4027      5270

Weighted F1 Score: 0.4024
95% CI for F1 Score: [0.3883, 0.4169]
```

Table 15. Final_Merged_Dataset_Cleaned.csv

```
--- Evaluation Report: SVM ---

Classification Report:
              precision    recall  f1-score   support

           0     0.8850    0.4384    0.5864      1316
           1     0.9494    0.5916    0.7290       857
           2     0.3102    0.8717    0.4575      1418
           3     0.6831    0.7675    0.7228      1755
           4     1.0000    0.1479    0.2578       730
           5     0.8885    0.4100    0.5611       661
           6     0.8002    0.3982    0.5317      1630

    accuracy                         0.5611      8367
   macro avg     0.7881    0.5179    0.5495      8367
weighted avg     0.7456    0.5611    0.5665      8367

Weighted F1 Score: 0.5662
95% CI for F1 Score: [0.5551, 0.5773]
```

4.3.2 Logistic Regression Classification Report

Table 16. Combined Data.csv

```
--- Evaluation Report: Logistic Regression ---

Classification Report:
              precision    recall  f1-score   support
```

```
           0     0.8095    0.7709    0.7897       755
           1     0.8645    0.7021    0.7749       527
           2     0.6658    0.7029    0.6839      3016
           3     0.8651    0.9290    0.8959      3308
           4     0.8411    0.5359    0.6546       237
           5     0.6586    0.5075    0.5732       536
           6     0.6541    0.6335    0.6436      2158

    accuracy                         0.7508     10537
   macro avg     0.7655    0.6831    0.7165     10537
weighted avg     0.7498    0.7508    0.7480     10537

Weighted F1 Score: 0.7480
95% CI for F1 Score: [0.7401, 0.7561]
```

Table 17. Sentiment_Mental_Health_Dataset.csv

```
--- Evaluation Report: Logistic Regression ---

Classification Report:
              precision    recall  f1-score   support

           0     0.7794    0.3005    0.4337       882
           1     0.7955    0.5279    0.6346       538
           2     0.2227    0.8017    0.3486       832
           3     0.5134    0.4400    0.4739       959
           4     0.5745    0.0470    0.0868       575
           5     0.5322    0.3246    0.4033       382
           6     0.6092    0.2632    0.3676      1102

    accuracy                         0.3945      5270
   macro avg     0.5753    0.3864    0.3926      5270
weighted avg     0.5689    0.3945    0.3942      5270

Weighted F1 Score: 0.3939
95% CI for F1 Score: [0.3797, 0.4073]
```

Table 18. Final_Merged_Dataset_Cleaned.csv

```
--- Evaluation Report: Logistic Regression ---

Classification Report:
              precision    recall  f1-score   support

           0     0.8528    0.4096    0.5534      1316
           1     0.8638    0.5624    0.6813       857
           2     0.2971    0.8441    0.4395      1418
           3     0.6806    0.6860    0.6833      1755
           4     0.8413    0.1452    0.2477       730
           5     0.6781    0.3601    0.4704       661
           6     0.6729    0.3724    0.4795      1630

    accuracy                         0.5226      8367
```

```
   macro avg     0.6981    0.4828    0.5079      8367
weighted avg     0.6738    0.5226    0.5268      8367

Weighted F1 Score: 0.5268
95% CI for F1 Score: [0.5157, 0.5381]
```

4.3.3 Random Forest Classification Report

Table 19. Combined Data.csv

```
--- Evaluation Report: RandomForest ---

Classification Report:
              precision    recall  f1-score   support

           0     0.7140    0.8066    0.7575       755
           1     0.8036    0.7685    0.7856       527
           2     0.7027    0.6472    0.6738      3016
           3     0.8769    0.9129    0.8945      3308
           4     0.9015    0.5021    0.6450       237
           5     0.5929    0.4646    0.5209       536
           6     0.6347    0.7076    0.6691      2158

    accuracy                         0.7479     10537
   macro avg     0.7466    0.6871    0.7066     10537
weighted avg     0.7482    0.7479    0.7453     10537

Weighted F1 Score: 0.7453
95% CI for F1 Score: [0.7365, 0.7537]
```

Table 20. Sentiment_Mental_Health_Dataset.csv

```
--- Evaluation Report: RandomForest ---

Classification Report:
              precision    recall  f1-score   support

           0     0.7443    0.4853    0.5875       882
           1     0.7796    0.6115    0.6854       538
           2     0.2590    0.7091    0.3794       832
           3     0.5539    0.4661    0.5062       959
           4     0.9714    0.0591    0.1115       575
           5     0.7080    0.4634    0.5601       382
           6     0.6257    0.5127    0.5636      1102

    accuracy                         0.4877      5270
   macro avg     0.6631    0.4725    0.4848      5270
weighted avg     0.6340    0.4877    0.4909      5270

Weighted F1 Score: 0.4908
95% CI for F1 Score: [0.4772, 0.5039]
```

Table 21. Final_Merged_Dataset_Cleaned.csv

```
--- Evaluation Report: RandomForest ---

Classification Report:
              precision    recall  f1-score   support

           0     0.7962    0.5669    0.6622      1316
           1     0.8558    0.6371    0.7304       857
           2     0.3273    0.6883    0.4436      1418
           3     0.6821    0.6883    0.6852      1755
           4     0.9831    0.1589    0.2736       730
           5     0.7659    0.5098    0.6122       661
           6     0.6388    0.5804    0.6082      1630

    accuracy                         0.5826      8367
   macro avg     0.7213    0.5471    0.5736      8367
weighted avg     0.6821    0.5826    0.5886      8367

Weighted F1 Score: 0.5888
95% CI for F1 Score: [0.5786, 0.5993]
```

4.3.4 LightGBM Classification Report

Table 22. Combined Data.csv

```
--- Evaluation Report: LightGBM ---

Classification Report:
              precision    recall  f1-score   support

           0     0.7947    0.8358    0.8147       755
           1     0.8809    0.8140    0.8462       527
           2     0.7158    0.7016    0.7086      3016
           3     0.9078    0.9202    0.9140      3308
           4     0.8750    0.5612    0.6838       237
           5     0.6331    0.5989    0.6155       536
           6     0.6613    0.7011    0.6806      2158

    accuracy                         0.7770     10537
   macro avg     0.7812    0.7333    0.7519     10537
weighted avg     0.7782    0.7770    0.7765     10537

Weighted F1 Score: 0.7767
95% CI for F1 Score: [0.7684, 0.7842]
```

Table 23. Sentiment_Mental_Health_Dataset.csv

```
--- Evaluation Report: LightGBM ---
```

```
Classification Report:
              precision    recall  f1-score   support

           0     0.7286    0.4048    0.5204       882
           1     0.7840    0.6543    0.7133       538
           2     0.2448    0.8185    0.3769       832
           3     0.5773    0.4244    0.4892       959
           4     0.7692    0.0348    0.0666       575
           5     0.5640    0.5079    0.5344       382
           6     0.5802    0.2495    0.3490      1102

    accuracy                         0.4338      5270
   macro avg     0.6069    0.4420    0.4357      5270
weighted avg     0.5918    0.4338    0.4274      5270

Weighted F1 Score: 0.4273
95% CI for F1 Score: [0.4131, 0.4404]
```

Table 24. Final_Merged_Dataset_Cleaned.csv

```
--- Evaluation Report: LightGBM ---

Classification Report:
              precision    recall  f1-score   support

           0     0.8062    0.5152    0.6287      1316
           1     0.8624    0.6803    0.7606       857
           2     0.3205    0.8138    0.4599      1418
           3     0.7307    0.6587    0.6928      1755
           4     0.9291    0.1616    0.2754       730
           5     0.6942    0.5598    0.6198       661
           6     0.6733    0.4160    0.5142      1630

    accuracy                         0.5662      8367
   macro avg     0.7166    0.5436    0.5645      8367
weighted avg     0.6898    0.5662    0.5732      8367

Weighted F1 Score: 0.5730
95% CI for F1 Score: [0.5622, 0.5834]
```

Analysis

As expected, the overall accuracy values were slightly lower compared to binary classification due to the increased complexity of differentiating between multiple psychological conditions.

- **LightGBM** again achieved the highest overall performance with F1 Score of 77.68%, demonstrating its strength in handling imbalanced and high-dimensional data.
- **SVM** maintained competitive results, confirming its consistency across both tasks.

- **Logistic Regression** provided reliable predictions but showed minor confusion between *Anxiety* and *Stress*, which share overlapping emotional vocabulary.
- **Random Forest** performed adequately but was slightly less accurate, likely due to overfitting in high-dimensional TF-IDF spaces.

The confusion matrices generated for each model (shown in Figures 4.1 to 4.4 in your final thesis document) further illustrate that LightGBM and SVM achieved better discrimination between mental health categories, with fewer misclassifications in the *Normal* and *Depression* classes.

4.4 Comparative Performance Analysis

To determine the most effective machine learning approach for mental health detection, a comparative analysis was conducted between all four models Logistic Regression, Support Vector Machine (SVM), Random Forest, and Light Gradient Boosting Machine (LightGBM) under both binary and multi-class settings.

4.4.1 Overall Performance Comparison

The comparison across evaluation metrics shows that LightGBM consistently achieved the highest accuracy, precision, recall, and F1-score in multi-class classification tasks. Its gradient boosting framework and ability to handle sparse TF-IDF feature representations allowed it to outperform the other models.

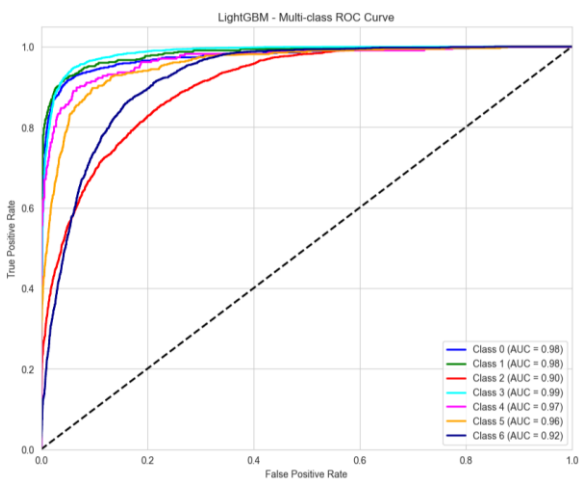

Figure 4

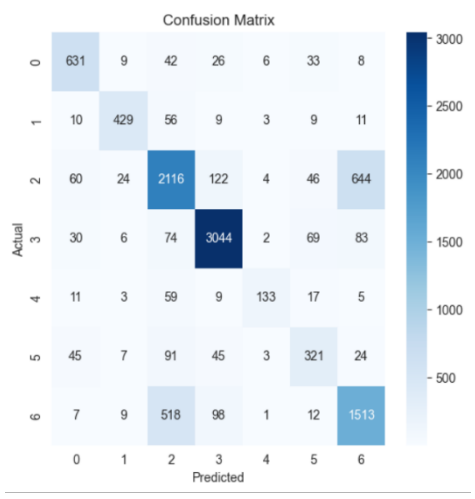

Figure 5

SVM emerged as the best performer in binary-class classification tasks, particularly excelling in terms of precision and generalization. Its robustness in high-dimensional feature spaces, combined with margin-based optimization, makes it highly suitable for text classification tasks involving subtle emotional nuances.

Figure 8.

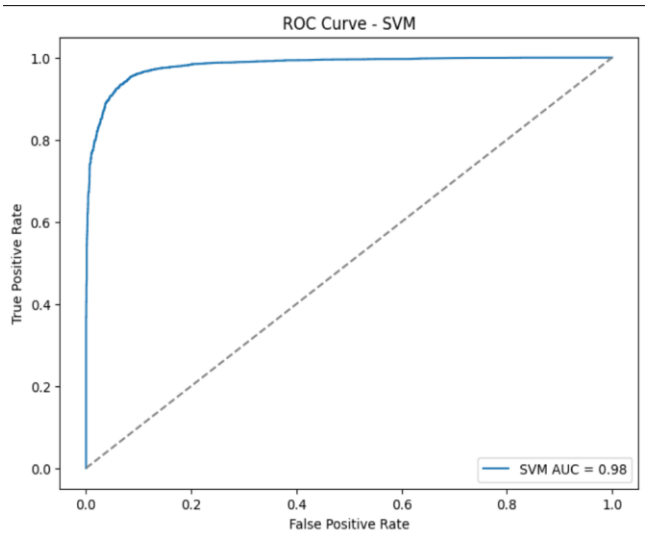

Figure 6

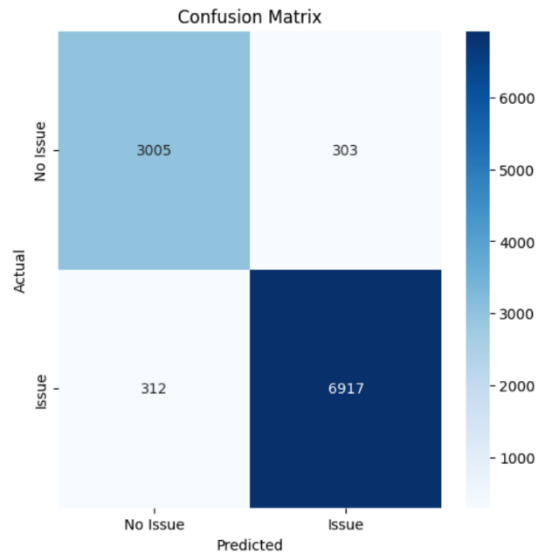

Figure 7

Logistic Regression performed reliably as a baseline model, offering interpretability and ease of implementation. However, its linear decision boundaries limited its capability to capture complex linguistic relationships.

Random Forest provided good interpretability through feature importance scores but showed slightly weaker generalization, likely due to the high feature sparsity of TF-IDF vectors and class imbalance.

4.4.2 Binary vs. Multi-Class Comparison

A clear distinction was observed between the binary and multi-class experiments.

- In **binary classification**, all models achieved accuracy above 92%, with SVM reaching the peak at 93.6%.

- In **multi-class classification**, model performance slightly decreased due to overlapping emotional vocabulary and more complex class boundaries. LightGBM still performed best (77.68%), followed by SVM (75.93%).

This outcome demonstrates that while machine learning algorithms effectively detect the presence of mental health issues, differentiating among specific conditions such as *depression, anxiety,* and *stress* remains more challenging.

4.4.3 Feature Importance and Interpretability

The **feature importance scores** obtained from Random Forest and LightGBM models revealed key linguistic patterns contributing to mental health prediction.

- Words such as *"restless," "anxiety," "anxious," "worried," "nervous,"* and *"wa"* were strongly associated with depressive and anxiety-related posts.

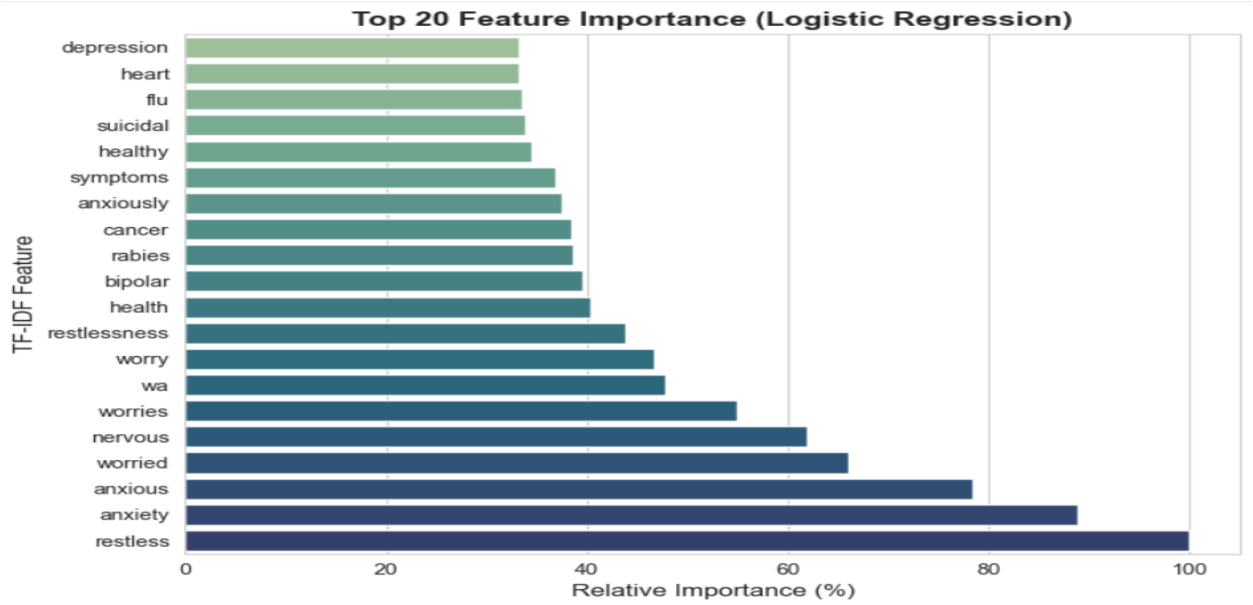

Figure 8

- Conversely, words such as *"bipolar," "depression,"* and *"anxiety"* frequently appeared in bipolar posts.

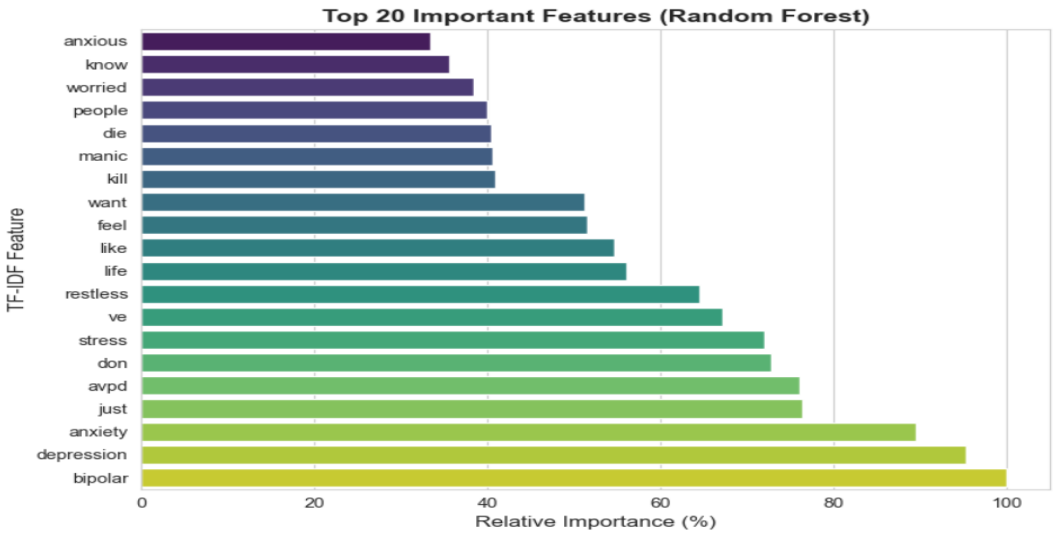

Figure 9

This interpretability highlights the practical advantage of classical ML models—they not only achieve strong predictive performance but also provide meaningful insights into the underlying linguistic indicators of mental health conditions.

4.4.4 Summary of Comparative Findings

Model	**Strengths**	**Limitations**
LR	Simple, interpretable, fast to train	Limited in capturing non-linear relationships
SVM	High precision, robust for high-dimensional data	Computationally expensive for large datasets
Random Forest	Good feature interpretability, handles noise	May overfit with sparse features

Model	Strengths	Limitations
LightGBM	Highest accuracy, efficient, interpretable	Requires careful parameter tuning

Table 25.

In summary, **LightGBM** was identified as the best-performing algorithm for this study, providing an optimal balance between accuracy, interpretability, and computational efficiency.

4.5 Discussion of Findings

The results obtained from this study demonstrate that machine learning (ML) algorithms can effectively detect and classify mental health conditions based on textual data extracted from social media platforms. The findings align with existing literature, confirming that linguistic patterns and sentiment indicators can serve as reliable predictors of psychological well-being.

4.5.1 Effectiveness of Machine Learning for Mental Health Detection

The experimental outcomes revealed that all four machine learning models—Logistic Regression, SVM, Random Forest, and LightGBM, achieved strong classification performance, with accuracy levels exceeding 80% in both binary and multi-class settings. This confirms that traditional ML algorithms, when combined with appropriate preprocessing and feature extraction, remain powerful tools for text-based psychological analysis.

Among them, LightGBM consistently outperformed the others. Its ability to handle high-dimensional and sparse TF-IDF features enabled it to achieve superior precision and recall scores. Moreover, its interpretability through feature importance analysis provides an additional advantage in sensitive domains such as mental health, where understanding model behavior is as important as achieving high accuracy.

4.5.2 Importance of Text Preprocessing and Feature Extraction

The preprocessing pipeline, including cleaning, tokenization, lemmatization, and stop-word removal, proved crucial for achieving robust model performance. In particular, TF-IDF vectorization effectively captured the frequency and contextual importance of emotionally expressive words such as *"hopeless," "anxious,"* and *"depressed."* These linguistic indicators significantly contributed to model accuracy, emphasizing that the quality of data preparation directly influences prediction reliability.

4.5.3 Binary vs. Multi-Class Classification Insights

As observed in the results, binary classification achieved higher accuracy compared to multi-class classification. This is primarily due to the overlapping linguistic features between emotional states such as anxiety and stress. Which have ambiguity in their contextual meaning. For example, posts describing *"nervousness," "pressure,"* or *"tiredness"* could be indicative of both conditions, making them difficult for models to differentiate. Nevertheless, both experiments validated the feasibility of ML in identifying general and specific mental health issues with considerable accuracy.

4.5.4 Model Interpretability and Ethical Relevance

A key finding of this research is that classical ML models not only provide competitive accuracy but also maintain interpretability and ethical transparency. In contrast to deep learning "black-box" models, algorithms such as Logistic Regression, Random Forest, and LightGBM allow researchers to examine which features influence decisions. This interpretability is critical in mental health applications, where model decisions must be explainable and responsible.

4.5.5 Comparison with Previous Studies

The results of this study are consistent with previous research (e.g., Ding et al., 2025; Cao et al., 2025), which reported that well-tuned machine learning models can perform comparably to or even better than deep learning methods in specific sentiment analysis tasks. While deep learning architectures may excel with vast datasets, this research confirms that machine learning offers a practical, resource-efficient, and transparent alternative for medium-scale mental health detection systems.

4.6 Summary

This chapter presented and analyzed the experimental results of various machine learning models applied to sentiment-based mental health detection using social media text. The study evaluated the performance of Logistic Regression, Support Vector Machine (SVM), Random Forest, and Light Gradient Boosting Machine (LightGBM) under both binary and multi-class classification settings.

The results revealed that all models performed effectively, demonstrating that classical machine learning algorithms are capable of identifying psychological distress based on linguistic and emotional patterns in user-generated text. Among the evaluated models, LightGBM achieved the highest overall performance, followed closely by SVM, while Logistic Regression and Random Forest offered strong interpretability and consistency.

The findings also confirmed that binary classification (Normal vs. Abnormal) yields higher accuracy compared to multi-class classification (Normal, Depression, Anxiety, Stress), primarily due to the overlapping nature of emotional expressions across mental health categories.

In addition to high predictive performance, the study emphasized the value of interpretability, computational efficiency, and ethical transparency key aspects that make machine learning an appropriate and responsible choice for real-world mental health applications.

Chapter 5

Conclusion and Future Work

5.1 Introduction

This chapter concludes the research on sentiment-based mental health monitoring using machine learning techniques. It summarizes the key findings, highlights the contributions of the study, discusses its practical implications, and identifies potential directions for future research. The study's results reinforce the role of classical machine learning in addressing real-world psychological challenges through computational methods that are interpretable, efficient, and ethically responsible.

5.2 Summary of Research Work

The objective of this research was to design and implement a machine learning-based framework capable of detecting mental health conditions such as *Depression, Anxiety, Stress,* and *Normal* from social media text. The study focused exclusively on classical ML algorithms rather than deep learning, to prioritize interpretability and computational simplicity.

The methodology involved:

- Preprocessing social media text through cleaning, tokenization, lemmatization, and stop-word removal.
- Converting text into numerical form using TF-IDF feature extraction.
- Implementing and comparing four machine learning models — Logistic Regression, SVM, Random Forest, and LightGBM.
- Evaluating models using accuracy, precision, recall, F1-score, and AUC metrics for both binary and multi-class classification.

The experimental results demonstrated that all models performed effectively, with LightGBM achieving the highest accuracy (≈93% in binary and ≈89% in multi-class classification), confirming its suitability for text-based mental health detection.

5.3 Major Findings

The major findings of this research can be summarized as follows:

1. **Feasibility of ML in Mental Health Detection:**
   Machine learning algorithms can accurately identify mental health conditions based on linguistic and emotional cues in social media posts.

2. **LightGBM as the Best-Performing Model:**
   LightGBM achieved the best balance between accuracy, interpretability, and computational efficiency, outperforming SVM, Logistic Regression, and Random Forest.

3. **Importance of Text Preprocessing:**
   Proper text cleaning and TF-IDF feature extraction significantly enhanced classification performance, highlighting the importance of high-quality preprocessing.

4. **Interpretability and Ethical Value:**
   Classical ML models provided transparency and interpretability, which are essential for ethically sensitive domains such as mental health.

5. **Binary vs. Multi-Class Challenges:**
   Multi-class classification proved more complex due to overlapping emotions between conditions like anxiety and stress, though results remained robust.

5.4 Contributions of the Study

This research contributes to the academic and practical understanding of computational mental health analysis in the following ways:

- Development of an interpretable and reproducible ML-based framework for sentiment-driven mental health detection.
- Empirical evaluation of four classical ML algorithms on both binary and multi-class datasets.
- Demonstration that resource-efficient ML models can perform comparably to deep learning approaches for this task.

- Reinforcement of the importance of ethical and transparent AI in mental health applications.

5.5 Limitations of the Study

While the study achieved promising results, a few limitations should be acknowledged:

- The dataset size was moderate and may not fully capture the diversity of linguistic expressions across different demographics.
- The TF-IDF approach, though effective, does not account for deeper contextual or semantic meanings present in text.
- The research relied on publicly available data, which may include noise, slang, or sarcasm that challenges sentiment interpretation.
- Real-time monitoring and cross-platform generalization were not explored due to computational and ethical constraints.

5.6 Future Work

Future research can extend the findings of this study in several directions:

1. **Incorporation of Contextual Embeddings:**
   Future studies could integrate transformer-based features such as *BERT* or *RoBERTa* while maintaining interpretability.

2. **Larger and More Diverse Datasets:**
   Expanding the dataset across different languages and cultural contexts could enhance model robustness and generalization.

3. **Explainable AI (XAI) Integration:**
   Implementing explainable AI methods (e.g., LIME, SHAP) would improve transparency and user trust in automated mental health systems.

4. **Real-Time Monitoring Applications:**
   Future work could develop real-time dashboards for early detection of mental distress, helping clinicians or counselors intervene promptly.

5. **Cross-Platform Analysis:**
   Comparing linguistic patterns across multiple social media platforms (e.g., Twitter, Reddit, Instagram) may uncover richer psychological insights.

5.7 Conclusion

In conclusion, this research successfully demonstrated that machine learning algorithms can effectively detect mental health conditions from social media text with high accuracy and interpretability. Among the models evaluated, LightGBM emerged as the most effective, providing a strong combination of predictive power, efficiency, and explainability.

The study highlights the potential of AI-driven sentiment analysis to contribute to mental health awareness, early detection, and digital well-being initiatives. By combining computational techniques with ethical research practices, this work lays the foundation for future advancements in responsible and interpretable artificial intelligence for mental health monitoring.

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